SUMMER OF MACHINE LEARNING AT SKOLTECH

LLM for Business Content Creation

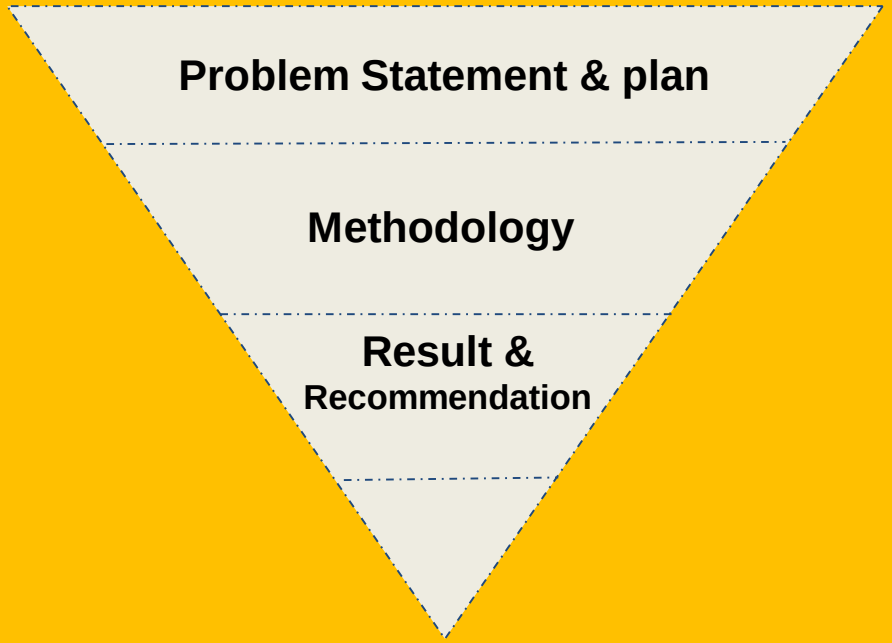
Team:

Dinar Galimov, Harbin Institute of Technology
Abdulrahman Nabil Abdo Al-sameai, Yangzhou University
Al-haimady Osama Yahya Mohammed Ahmed, Yangzhou University,
Umar Muhammad Mustapha Kumshe, Hohai University

Curator(s) name: Dmitry Bazyukin, Dmitry Akulichev, Alexey Chernyshov, Andrey Kostornoy, Artem Kopin

Skoltech Applied AI Center

Bank of Russia



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Problem Statement & plan

- Defining the Challenge: The complexity of central bank language and its impact on public trust.
- Our Plan: To use a lightweight, fine-tuned LLM to simplify these communications without losing meaning.

Methodology

- Our Technical Approach: The process of fine-tuning the Vikhr-Llama model using Unsupervised Style Transfer.
- Overcoming Hurdles: How we navigated hardware limitations, dependency issues, and data integrity problems.

Result & Recommendation

- Our Findings: Presenting clear, "before-and-after" examples of simplified text and the measurable improvement in key metrics.
- Next Steps: Proposing actionable recommendations for integration and future research.



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The Challenge & Our Solution
Enhancing Central Bank Communication with AI

The Problem: Official communications from central banks are often filled with technical jargon and complex language, making them inaccessible to the public.

Why It's Important: This lack of clarity can undermine public trust and weaken the impact of monetary policy.

Our Solution: We use a compact and efficient Large Language Model (LLM) to simplify complex regulatory language into clear, accessible Russian.

Model: Vikhr-Llama-3.2-1B-instruct, a lightweight model ideal for real-world deployment.

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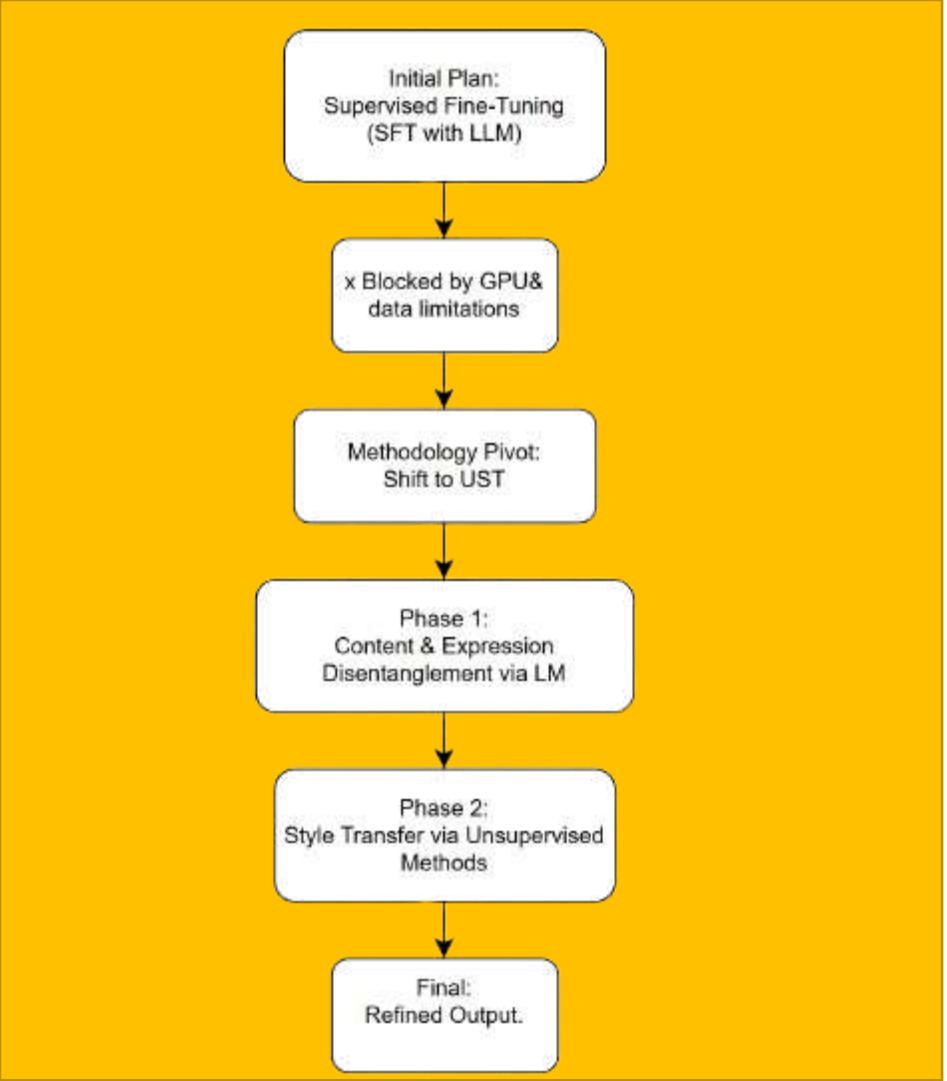
The Challenge & Our Solution
Enhancing Central Bank Communication with AI



Our Process: We transform dense text into clear communication through a rigorous funnel:

Simplify with LLM

- Evaluate Transparency (AlinaEstimator)
- Evaluate Readability (ARIEstimator)
- Expert Review



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Methodology: From Roadblock to Breakthrough

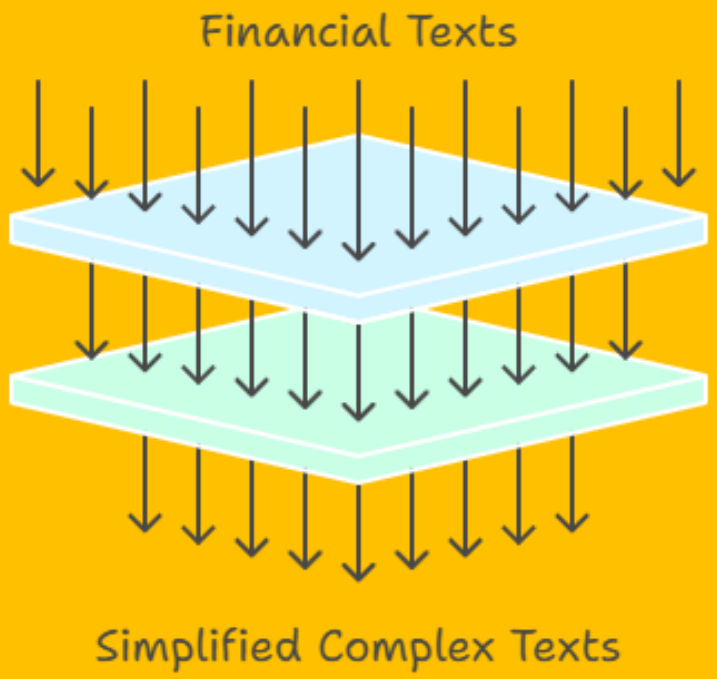
How We Trained the Model

Initial Goal: Supervised Fine-Tuning (SFT)

The standard approach, SFT, requires a paired dataset of "complex" and "simple" texts to teach the model a direct transformation.

The Roadblock: The Synthetic Data Bottleneck

Creating the required paired dataset was infeasible due to resource and access restrictions in our development environment.



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Methodology: From Roadblock to Breakthrough

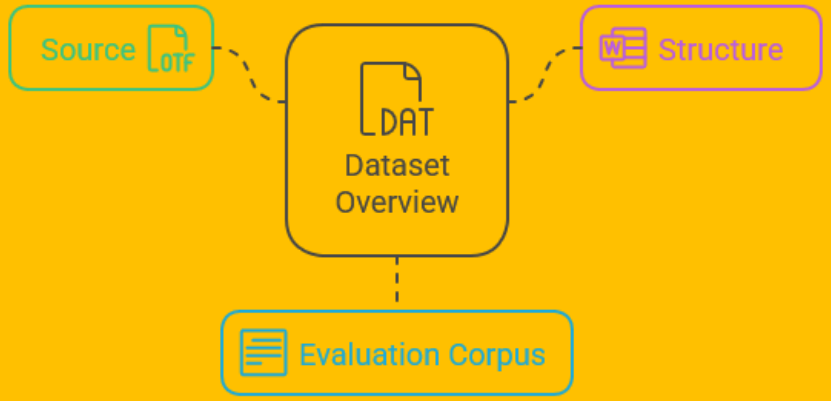
The Pivot: Unsupervised Style Transfer (UST)

Faced with constraints, we adopted a more practical approach: Unsupervised Style Transfer.

Phase 1: Style-Tuning: We fine-tuned the model exclusively on a dataset of simple, clear financial texts. This imbued the model with a strong stylistic preference for clarity.

Phase 2: Prompt-Based Inference: We then used strong instructional prompts to guide the "style-tuned" model to rewrite complex texts in the simple style it had learned.

Overcoming Hurdles: This process required solving significant technical challenges related to hardware (NPU), outdated libraries, and data integrity.



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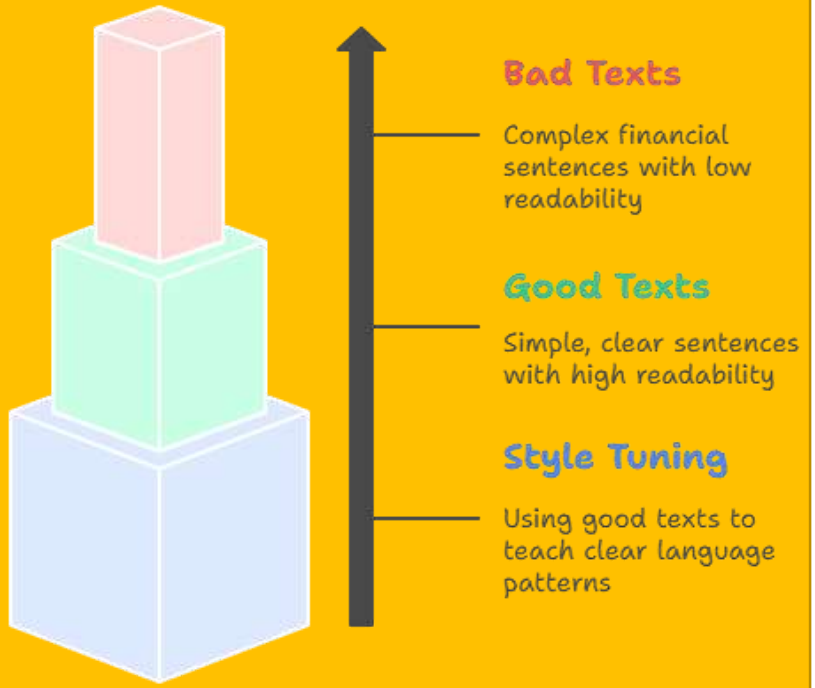
A Closer Look at the Datasets

Dataset Overview

Source: The input data consists of Russian-language financial texts from the Bank of Russia.

Structure: Each text entry includes a unique ID, pre-evaluated scores, the original text, and a word count.

Evaluation Corpus: The evaluation metrics were developed using a corpus of 10,000 Russian texts, which were classified into ten distinct readability levels.



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A Closer Look at the Datasets

Training Datasets for Fine-Tuning

Our methodology required separating the source data into two distinct, unpaired collections:

bad_texts.json: This collection contained complex financial sentences characterized by low readability scores.

good_texts.json: This collection consisted of simple, clear sentences with high readability scores.

Unsupervised Approach: Because the "bad" and "good" examples were not directly paired, a standard Supervised Fine-Tuning (SFT) was not possible. The

good_texts.json dataset was used exclusively in the "style-tuning" phase of our Unsupervised Style Transfer approach to teach the model the patterns of clear financial language.



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Results & Discussion

Measurable Success

Proven Success: Our Unsupervised Style Transfer method successfully simplified complex financial text while preserving all critical numerical data.

Example of a Successful Simplification:

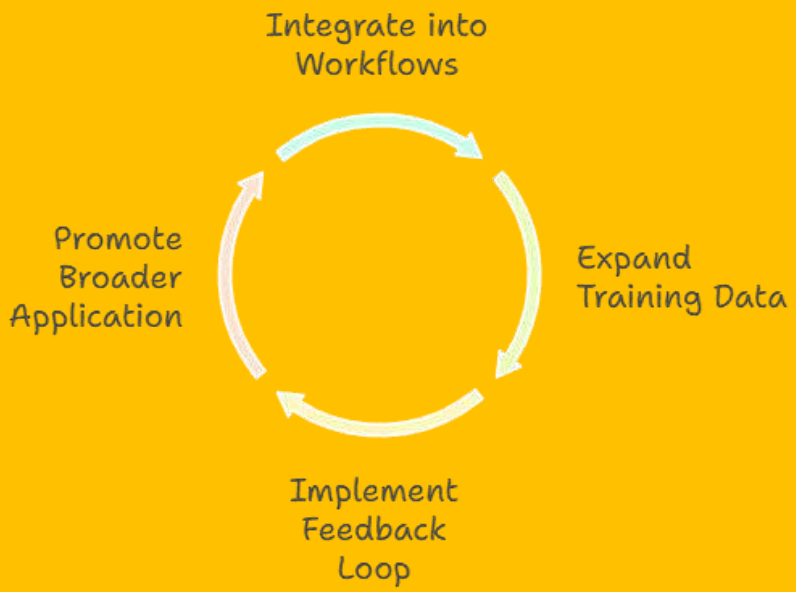
The model deconstructed a dense sentence into multiple, more digestible parts.

All key data (770 billion rubles, -14%, -5%, +8%) was preserved accurately.

Quantifiable Improvement: The simplified text showed significant improvement in our core metrics.

Transparency (AlinaEstimator): The score doubled from 3 to 6, indicating much clearer communication.

Readability (ARIEstimator): The score decreased from 6.4 to 6.1, meaning the text is easier to understand.



Recommendations

Actionable Recommendations

Integrate into Workflows: The Bank of Russia should consider integrating this model as an assistive tool to generate initial drafts of press releases and reports for human expert review.

Continuous Model Refinement: Improve the model by expanding the training dataset with a wider variety of historical documents and speech transcripts. A feedback loop using expert corrections should be implemented for ongoing fine-tuning.

Promote Broader Application: Other government agencies, both in Russia and internationally, could adapt this efficient and transferable methodology to improve their own communications.

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Future Research Directions



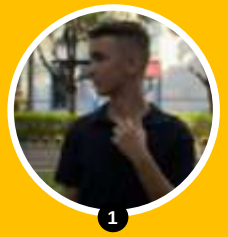
Quantitative Impact Assessment: Conduct surveys and A/B tests with diverse audiences to quantitatively measure the real-world impact of the simplified texts on public comprehension and trust.

Explore Multimodality: Future research should explore multimodal LLMs capable of interpreting and simplifying information from various formats like charts, graphs, and videos.

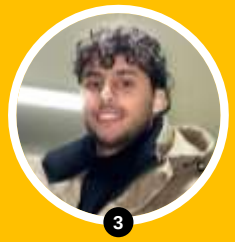
Develop Ethical Guardrails: Further research into explainability (XAI) is needed to help human overseers understand the model's choices, which builds trust and ensures responsible deployment.

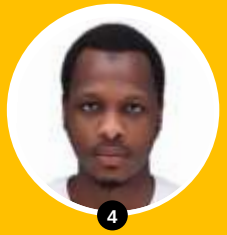
Bank of Russia

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Dinar Galimov

Affiliation: Harbin Institute of Technology

- Executed the testing phase, applying the AlinaEstimator and ARIEstimator metrics to measure the output quality.
- Performed final evaluation on simplified text outputs & Led literature review

Abdulrahman Nabil Abdo Al-sameai

Affiliation:Yangzhou University

- Managed complex/simple text dataset preparation
- Configured training environment; resolved NPU & software issues

Al-haimady Osama Yahya Mohammed Ahmed

Affiliation:Yangzhou University

- Fine-tuned Vikhr-Llama-3.2-1B-instruct using Unsupervised Style Transfer
- Implemented style-tuning, prompt-based inference, and resolved data integrity issues

Umar Muhammad Mustapha Kumshe

Affiliation:Hohai University

- Prepared final report, including limitations and future recommendations.
- Guided pivot to Unsupervised Style Transfer due to project constraints.



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Thank you !!

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